Assignment 2 - 1011 ICT

GROUP NUMBER: 3

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Declaration Except where appropriately acknowledged, this assignment is our own work, has been expressed in our own words and has not previously been submitted for assessment. We have also retained a copy of this assessment piece for our own records.			
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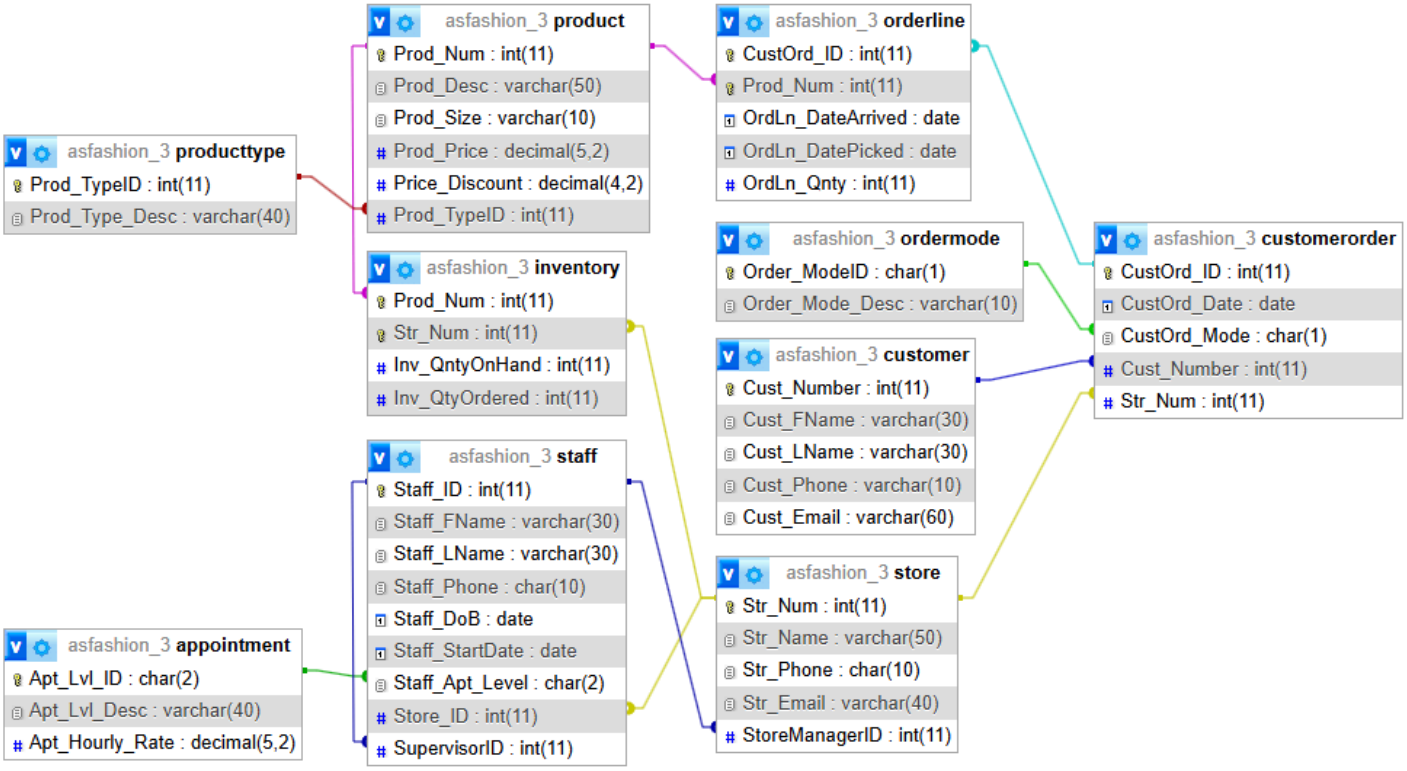
MySQL System Schema



SQL Queries for Database Creation in Task 1

```
DROP DATABASE IF EXISTS aSFashion_3; -- delete the previously existing database for
easy testing purposes
CREATE DATABASE aSFashion_3; -- create new database
USE aSFashion_3; -- put the following into the database

CREATE TABLE Customer(
    Cust_Number int PRIMARY KEY AUTO_INCREMENT NOT NULL, -- Primary Key, Auto
Increment, Not Null
    Cust_FName varchar(30), -- e.g Mohammad, John
    Cust_LName varchar(30), -- e.g Awrangjeb, Wang
    Cust_Phone varchar(10), -- Includes area code, e.g 0737355032
    Cust_Email varchar(60) -- e.g m.fraser@gmail.com
);

CREATE TABLE Store(
    Str_Num int PRIMARY KEY AUTO_INCREMENT NOT NULL,
    Str_Name varchar(50),
    Str_Phone char(10),
    Str_Email varchar(40),
    StoreManagerID int
);

CREATE TABLE Appointment (
    Apt_Lvl_ID CHAR(2) PRIMARY KEY,
    Apt_Lvl_Desc VARCHAR(40),
    Apt_Hourly_Rate DECIMAL(5, 2)
);

CREATE TABLE Staff (
    Staff_ID INT PRIMARY KEY AUTO_INCREMENT,
    Staff_FName VARCHAR(30),
    Staff_LName VARCHAR(30),
    Staff_Phone CHAR(10),
    Staff_DoB DATE,
    Staff_StartDate DATE,
    Staff_Apt_Level CHAR(2),
    Store_ID INT,
    SupervisorID INT,
    FOREIGN KEY (Staff_Apt_Level) REFERENCES Appointment(Apt_Lvl_ID),
    FOREIGN KEY (Store_ID) REFERENCES STORE(Str_Num),
    FOREIGN KEY (SupervisorID) REFERENCES Staff(Staff_ID)
);
```

```
ALTER TABLE Store
ADD CONSTRAINT fk
FOREIGN KEY (StoreManagerID) REFERENCES Staff(Staff_ID);

CREATE TABLE PRODUCTTYPE (
    Prod_TypeID INT PRIMARY KEY AUTO_INCREMENT,
    Prod_Type_Desc VARCHAR(40)
);

CREATE TABLE PRODUCT (
    Prod_Num INT PRIMARY KEY AUTO_INCREMENT,
    Prod_Desc VARCHAR(50),
    Prod_Size VARCHAR(10),
    Prod_Price DECIMAL(5, 2),
    Price_Discount DECIMAL(4, 2),
    Prod_TypeID INT,
    FOREIGN KEY (Prod_TypeID) REFERENCES PRODUCTTYPE(Prod_TypeID)
);

CREATE TABLE INVENTORY (
    Prod_Num INT,
    Str_Num INT,
    Inv_QntyOnHand INT,
    Inv_QtyOrdered INT,
    PRIMARY KEY (Prod_Num, Str_Num),
    FOREIGN KEY (Prod_Num) REFERENCES PRODUCT(Prod_Num),
    FOREIGN KEY (Str_Num) REFERENCES STORE(Str_Num)
);

CREATE TABLE ORDERMODE (
    Order_ModeID CHAR(1) PRIMARY KEY,
    Order_Mode_Desc VARCHAR(10)
);

CREATE TABLE CUSTOMERORDER (
    CustOrd_ID INT PRIMARY KEY AUTO_INCREMENT,
    CustOrd_Date DATE,
    CustOrd_Mode CHAR(1),
    Cust_Number INT,
    Str_Num INT,
    FOREIGN KEY (CustOrd_Mode) REFERENCES ORDERMODE(Order_ModeID),
    FOREIGN KEY (Cust_Number) REFERENCES CUSTOMER(Cust_Number),
    FOREIGN KEY (Str_Num) REFERENCES STORE(Str_Num)
);
```

```
CREATE TABLE ORDERLINE (
    CustOrd_ID INT,
    Prod_Num INT,
    OrdLn_DateArrived DATE,
    OrdLn_DatePicked DATE,
    OrdLn_Qnty INT,
    PRIMARY KEY (CustOrd_ID, Prod_Num),
    FOREIGN KEY (CustOrd_ID) REFERENCES CUSTOMERORDER(CustOrd_ID),
    FOREIGN KEY (Prod_Num) REFERENCES PRODUCT(Prod_Num)
);
```

SQL Queries for Data Insertion in Task 2

```
INSERT INTO customer (Cust_Number,Cust_FName,Cust_LName,Cust_Phone,Cust_Email)
VALUES
(NULL,'Rosemarie','Bradford',0437661489,'r.bradford@gmail.com'),
(NULL,'Cornell','Woodward',0359876246,'c.woodward@yahoo.com.au'),
(NULL,'Noe','Short',0219384752,'n.short@aol.com'),
(NULL,'Tammie','Norris',0849862457,'t.norris@live.com'),
(NULL,'Olin','Reilly',0694625791,'o.reilly@hotmail.com');

INSERT INTO appointment (Apt_Lvl_ID,Apt_Lvl_Desc,Apt_Hourly_Rate)
VALUES
('M1','Manager Level 1',110.50),
('M2','Manager Level 2',55.12),
('S1','Supervisor Level 1',25.40),
('S2','Supervisor Level 2',20.30),
('C1','Cashier Level 1',17.10);

INSERT INTO store (Str_Num,Str_Name,Str_Phone,Str_Email)
VALUES
(NULL,'a*Fashion Melbourne',0359642350,'melboune@asfashion.com.au'),
(NULL,'a*Fashion Sydney',0234851276,'sydney@asfashion.com.au'),
(NULL,'a*Fashion Perth',0836785146,'perth@asfashion.com.au'),
(NULL,'a*Fashion Adelaide',0871356784,'adelaide@asfashion.com.au'),
(NULL,'a*Fashion Brisbane',0713459764,'brisbane@asfashion.com.au');

INSERT INTO staff
(Staff_ID,Staff_FName,Staff_LName,Staff_Phone,Staff_DoB,Staff_StartDate,Staff_
Apt_Level,Store_ID)
VALUES
(NULL,'Huey','English',0468743198,'1989-10-08','2022-06-12','S1',1),
(NULL,'Gabrielle','Potts',0434865125,'1998-11-21','2021-09-18','M1',1),
(NULL,'Selma','Cobb',0442789653,'2001-04-05','2023-04-14','S2',2),
(NULL,'Mario','Garrison',0415346895,'1982-01-22','2024-01-03','C1',1),
(NULL,'Nolan','Raymond',0443245876,'2006-09-12','2024-12-20','C1',2);

UPDATE staff
SET SupervisorID = CASE
    WHEN Staff_ID IN (1, 2) THEN 2
    WHEN Staff_ID IN (3, 5) THEN 3
    WHEN Staff_ID = 4 THEN 1
    ELSE SupervisorID
```

```
END;

UPDATE store
SET StoreManagerID = 2
WHERE Str_Num IN (1, 2, 3, 4, 5);

INSERT INTO producttype (Prod_TypeID,Prod_Type_Desc)
VALUES
(NULL,'Shirt'),
(NULL,'Pant'),
(NULL,'Sock'),
(NULL,'Shoe'),
(NULL,'Hat');

INSERT INTO product
(Prod_Num,Prod_Desc,Prod_Size,Prod_Price,Price_Discount,Prod_TypeID)
VALUES
(NULL,'Shirt for man','Large',20.95,0.00,1),
(NULL,'Pant for woman','Medium',34.99,10.00,2),
(NULL,'Sock for child','Small',5.50,0.00,3),
(NULL,'Shoe for man','Medium',120.95,20.00,4),
(NULL,'Hat, unisex','Medium',15.40,5.00,5);

INSERT INTO inventory (Prod_Num,Str_Num,Inv_QntyOnHand,Inv_QtyOrdered)
VALUES
(1,1,700,120),
(2,1,1200,200),
(3,2,120,30),
(4,4,2000,300),
(5,5,360,80);

INSERT INTO ordermode (Order_ModeID,Order_Mode_Desc)
VALUES
('O','Online'),
('I','In store');

INSERT INTO customerorder
(CustOrd_ID,CustOrd_Date,CustOrd_Mode,Cust_Number,Str_Num)
VALUES
(NULL,'2024-06-12','O',1,1),
(NULL,'2023-07-24','I',2,2),
(NULL,'2024-03-19','I',4,3),
(NULL,'2022-02-08','O',5,4),
(NULL,'2024-03-04','O',3,5);
```

```
INSERT INTO orderline
(CustOrd_ID,Prod_Num,OrdLn_DateArrived,OrdLn_DatePicked,OrdLn_Qnty)
VALUES
(1,2,'2024-07-01','2024-07-12',2),
(2,5,'2023-08-03','2023-08-03',5),
(3,1,'2024-04-01','2024-04-03',10),
(4,3,'2022-02-14','2022-02-27',3),
(5,4,'2024-03-09','2024-03-15',8);
```

Reports on SQL Queries in Task 3

Query 1:

```
SELECT CONCAT(Staff_FNAME, ' ', Staff_LNAME) AS "Full Name", CONCAT('$',
Apt_Hourly_Rate) AS "Hourly Salary"
FROM Staff
JOIN Appointment ON Staff.Staff_Apt_Level = Appointment.Apt_Lvl_ID
ORDER BY Apt_Hourly_Rate DESC;
```

Output table:

Full Name	Hourly Salary
Gabrielle Potts	$110.50
Huey English	$25.40
Selma Cobb	$20.30
Mario Garrison	$17.10
Nolan Raymond	$17.10

Query 2:

```
SELECT CONCAT(Cust_FName, ' ', Cust_LName) AS "Full Name",
CUSTOMERORDER.CustOrd_Date AS "Order Date"
FROM CUSTOMER
JOIN CUSTOMERORDER ON CUSTOMERORDER.Cust_Number = CUSTOMER.Cust_Number
ORDER BY CustOrd_Date ASC
LIMIT 1 OFFSET 1;
```

Output table:

Full Name	Order Date
Cornell Woodward	2023-07-24

Query 3:

```
SELECT Str_Name AS "Store Name", CONCAT(Staff.Staff_FNAME, ' ', Staff.Staff_LNAME)
AS "Full Name",
Staff.Staff_Apt_Level AS "Appointment Level", CONCAT('$',
Appointment.Apt_Hourly_Rate) AS "Hourly Rate"
FROM Store
JOIN Staff ON Store.StoreManagerID = Staff.Staff_ID
JOIN Appointment ON Staff.Staff_Apt_Level = Appointment.Apt_Lvl_ID
WHERE Appointment.Apt_Lvl_Desc LIKE "Manager%"
ORDER BY Appointment.Apt_Hourly_Rate DESC;
```

Output table:

Store Name	Full Name	Appointment Level	Hourly Rate
a*Fashion Perth	Gabrielle Potts	M1	$110.50
a*Fashion Sydney	Gabrielle Potts	M1	$110.50
a*Fashion Melbourne	Gabrielle Potts	M1	$110.50
a*Fashion Brisbane	Gabrielle Potts	M1	$110.50
a*Fashion Adelaide	Gabrielle Potts	M1	$110.50

Query 4:

```
SELECT PRODUCT.Prod_Num AS "Product Number", Prod_Desc AS "Product",
PRODUCTTYPE.Prod_Type_Desc AS "Type", Prod_Size AS "Size", ORDERLINE.OrdLn_Qnty AS
"Total Qty. Sold"
FROM PRODUCT
JOIN PRODUCTTYPE ON PRODUCT.Prod_TypeID = PRODUCTTYPE.Prod_TypeID
JOIN ORDERLINE ON PRODUCT.Prod_Num = ORDERLINE.Prod_Num
WHERE ORDERLINE.OrdLn_DatePicked IS NOT NULL
ORDER BY ORDERLINE.OrdLn_Qnty DESC;
```

Output table:

Product Number	Product	Type	Size	Total Qty. Sold
1	Shirt for man	Shirt	Large	10
4	Shoe for man	Shoe	Medium	8
5	Hat, unisex	Hat	Medium	5
3	Sock for child	Sock	Small	3
2	Pant for woman	Pant	Medium	2

Query 5:

```
SELECT PRODUCT.Prod_Num AS "Product Number", Prod_Desc AS "Description",
PRODUCTTYPE.Prod_Type_Desc AS "Type", ORDERLINE.OrdLn_Qnty AS "Total Qty. Sold"
FROM PRODUCT
JOIN PRODUCTTYPE ON PRODUCT.Prod_TypeID = PRODUCTTYPE.Prod_TypeID
JOIN ORDERLINE ON PRODUCT.Prod_Num = ORDERLINE.Prod_Num
WHERE ORDERLINE.OrdLn_DatePicked <= '2024-05-20'
ORDER BY PRODUCT.Prod_Num ASC, ORDERLINE.OrdLn_Qnty ASC;
```

Output table:

Product Number	Description	Type	Total Qty. Sold
1	Shirt for man	Shirt	10
3	Sock for child	Sock	3
4	Shoe for man	Shoe	8
5	Hat, unisex	Hat	5